1. The hybridization of carbon in methane is

A) sp

B) sp^2

C) sp^3

D) dsp^2

Answer: C

---

2. Which has maximum bond angle?

A) NH_3

B) H_2O

C) CH_4

D) CO_2

Answer: D

---

3. The oxidation state of Mn in $KMnO_4$ is

A) +5

B) +6

C) +7

D) +4

Answer: C

---

4. Which has highest electronegativity?

A) O

B) F

C) Cl

D) N

Answer: B

---

5. Which compound shows resonance?

A) CH_4

B) NH_3

C) CO_3^{2-}

D) H_2O

Answer: C

---

6. Which is strongest acid?

A) HF

B) HCl

C) HBr

D) HI

Answer: D

---

7. Which has maximum boiling point?

A) CH_4

B) NH_3

C) H_2O

D) CO_2

Answer: C

---

8. Which law states PV = constant?

A) Boyle's law

B) Charles law

C) Avogadro law

D) Dalton law

Answer: A

---

9. Unit of rate constant for first order reaction is

A) s^{-1}

B) mol L^{-1}

C) L mol^{-1} s^{-1}

D) mol^2

Answer: A

10. Half-life of first order reaction is

A) Depends on concentration

B) Constant

C) Zero

D) Infinite

Answer: B

---

11. Which is a state function?

A) Work

B) Heat

C) Internal energy

D) Path

Answer: C

---

12. Entropy increases in

A) Freezing

B) Condensation

C) Evaporation

D) Solidification

Answer: C

---

13. Which is exothermic?

A) Melting

B) Boiling

C) Combustion

D) Sublimation

Answer: C

---

14. Which is strongest base?

A) NH_3

B) NaOH

C) KOH

D) $Ca(OH)_2$

Answer: C

---

15. pH of neutral solution is

A) 0

B) 7

C) 14

D) 1

Answer: B

---

16. Which is buffer solution?

A) HCl + NaCl

B) NH_3 + NH_4Cl

C) NaOH + NaCl

D) H_2SO_4 + Na_2SO_4

Answer: B

---

17. Which has maximum lattice energy?

A) NaCl

B) MgO

C) KCl

D) CaO

Answer: B

---

18. Which element has highest ionization energy?

A) Na

B) Mg

C) Al

D) Cl

Answer: D

---

19. Which is paramagnetic?

A) N_2

B) O_2

C) H_2

D) He

Answer: B

---

20. Which orbital has highest energy?

A) 2s

B) 2p

C) 3s

D) 3p

Answer: D

---

21. Which quantum number defines shape?

A) n

B) l

C) m

D) s

Answer: B

---

22. Which rule explains filling of orbitals?

A) Pauli

B) Hund

C) Aufbau

D) All

Answer: D

---

23. Which compound shows hydrogen bonding?

A) CH_4

B) NH_3

C) CO_2

D) CCl_4

Answer: B

---

24. Which is strongest oxidizing agent?

A) F_2

B) Cl_2

C) Br_2

D) I_2

Answer: A

---

25. Which is strongest reducing agent?

A) Li

B) Na

C) K

D) Cs

Answer: D

---

26. Which has highest atomic radius?

A) F

B) Cl

C) Br

D) I

Answer: D

---

27. Which is amphoteric oxide?

A) Na_2O

B) Al_2O_3

C) CO_2

D) SO_2

Answer: B

---

28. Which is strongest acid among oxyacids?

A) HClO

B) $HClO_2$

C) $HClO_3$

D) $HClO_4$

Answer: D

---

29. Which has highest bond energy?

A) F_2

B) Cl_2

C) Br_2

D) I_2

Answer: B

---

30. Which shows maximum catenation?

A) C

B) Si

C) Ge

D) Sn

Answer: A

---

31. Which functional group is aldehyde?

A) -COOH

B) -CHO

C) -OH

D) -CO-

Answer: B

---

32. Which is ketone?

A) CH_3CHO

B) CH_3COCH_3

C) CH_3OH

D) HCOOH

Answer: B

---

33. Which is carboxylic acid?

A) CH_3OH

B) CH_3COOH

C) CH_3CHO

D) CH_4

Answer: B

---

34. Which reaction is addition?

A) Hydrogenation

B) Combustion

C) Substitution

D) Elimination

Answer: A

---

35. Which reaction is substitution?

A) $CH_4 + Cl_2$

B) Alkene + H_2

C) Dehydration

D) Oxidation

Answer: A

---

36. Which is aromatic?

A) Benzene

B) Ethane

C) Ethene

D) Propane

Answer: A

---

37. Huckel rule is

A) 4n

B) 4n+2

C) 2n

D) n^2

Answer: B

---

38. Which is nucleophile?

A) NH_3

B) H^+

C) BF_3

D) $AlCl_3$

Answer: A

---

39. Which is electrophile?

A) OH^-

B) NH_3

C) NO_2^+

D) Cl^-

Answer: C

---

40. Which solvent is polar?

A) Benzene

B) Water

C) Hexane

D) Ether

Answer: B

41. Which is ideal gas condition?

A) High pressure

B) Low temperature

C) Low pressure

D) High density

Answer: C

---

42. Van der Waals equation corrects

A) Volume & pressure

B) Temperature

C) Energy

D) Density

Answer: A

---

43. Colligative properties depend on

A) Nature

B) Number of particles

C) Mass

D) Volume

Answer: B

---

44. Which has highest boiling point elevation?

A) NaCl

B) Glucose

C) Urea

D) KNO_3

Answer: A

---

45. Which is electrolyte?

A) Sugar

B) NaCl

C) Urea

D) Alcohol

Answer: B

---

46. Conductivity increases with

A) Temperature

B) Pressure

C) Volume

D) Density

Answer: A

---

47. Which is corrosion?

A) Oxidation

B) Reduction

C) Hydrolysis

D) Neutralization

Answer: A

---

48. Rusting requires

A) O_2

B) H_2O

C) Both

D) None

Answer: C

---

49. Which metal is least reactive?

A) Na

B) Fe

C) Cu

D) K

Answer: C

---

50. Which is noble gas?

A) N

B) O

C) Ne

D) Cl

Answer: C

---

51. Which has zero valency?

A) He

B) H

C) O

D) N

Answer: A

---

52. Which is hardest allotrope of carbon?

A) Graphite

B) Diamond

C) Fullerene

D) Coal

Answer: B

---

53. Which is conductor?

A) Diamond

B) Graphite

C) Plastic

D) Wood

Answer: B

---

54. Which is strongest intermolecular force?

A) London

B) Dipole

C) Hydrogen bonding

D) Ionic

Answer: D

---

55. Which is fastest reaction?

A) Ionic

B) Covalent

C) Redox

D) Slow

Answer: A

---

56. Activation energy is minimum for

A) Fast reaction

B) Slow reaction

C) Endothermic

D) None

Answer: A

---

57. Catalyst changes

A) Rate

B) Equilibrium

C) Energy

D) Products

Answer: A

---

58. Le Chatelier principle applies to

A) Equilibrium

B) Kinetics

C) Thermodynamics

D) Electrochemistry

Answer: A

---

59. Which increases equilibrium constant?

A) Temperature

B) Pressure

C) Volume

D) Catalyst

Answer: A

---

60. Gibbs free energy is

A) G = H - TS

B) H + TS

C) TS - H

D) H/T

Answer: A